

Part VI

Constrained Optimization II

Lecture Outline

- ① Lagrange Multipliers
- ② First order necessary conditions
- ③ Second order necessary conditions
- ④ Second order sufficient conditions

Lagrange Multipliers

- Lagrange multipliers are scalars $\lambda_1, \dots, \lambda_p$ and $\mu_1, \dots, \mu_q$, with $\mu_i \geq 0$, used to relate constrained optimization problems to unconstrained optimization problems.
- If x' is a minimizer for

$$\min f(x) \quad \text{s.t.} \quad A(x) = 0$$

then $\nabla f(x') = \sum \lambda_i \nabla a_i(x')$.

- If x' is a minimizer for

$$\min f(x) \quad \text{s.t.} \quad C(x) \geq_e 0$$

then $\nabla f(x') = \sum \mu_i \nabla c_i(x')$, with $\mu_i = 0$ if $c_i(x') > 0$.

Tangent and normal planes

Technical remark: Let there be given a C^1 function

$$h : \mathbb{R}^n \rightarrow \mathbb{R}^m; \ x \mapsto h(x) = [h_1(x) \cdots h_m(x)]^\top,$$

and assume that the Jacobian

$$J_h(x) = [\nabla h_1(x) \cdots \nabla h_m(x)]^\top \in \mathbb{R}^{m \times n},$$

has constant rank k in an open neighborhood of the point $x' \in \mathbb{R}^n$.

The tangent plane $T_{x'} h$ at $x' \in \mathbb{R}^n$ to the contour surface (or level set)

$$C_{x'} h = \{x \in \mathbb{R}^n \mid h(x) = h(x')\},$$

of h through x' is the $(n - k)$ -dimensional hyperplane

$$T_{x'} h = \{x \in \mathbb{R}^n \mid J_h(x')(x - x') = 0\}.$$

The normal plane $N_{x'} h$ to $T_{x'} h$ is the k -dimensional hyperplane

$$N_{x'} h = \text{range}(J(x')^\top) + x' = \text{span}\{\nabla h_i(x)\} + x'.$$

Note that for $m = 1$

$$T_{x'} h = \{x \in \mathbb{R}^n \mid \nabla h(x')^\top (x - x') = 0\},$$

$$N_{x'} h = \{x \in \mathbb{R}^n \mid x = \alpha \nabla h(x') + x', \ \alpha \in \mathbb{R}\}.$$

Tangent and normal planes connected to optimization

Consider

$$\min f(x) \quad \text{s.t.} \quad \mathbf{a}(x) = 0,$$

and let $x' \in \mathbb{R}^n$ be regular.

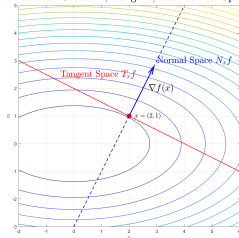
The tangent planes

$$T_{x', f} = \{x \in \mathbb{R}^n \mid \nabla f(x')^\top (x - x') = 0\},$$

$$T_{x', \mathbf{a}} = \{x \in \mathbb{R}^n \mid J_{\mathbf{a}}(x')(x - x') = 0\}, \quad J_{\mathbf{a}}(x') = [\nabla a_1(x') \cdots \nabla a_p(x')]^\top \in \mathbb{R}^{p \times n},$$

are then hyperplanes in $\mathbb{R}^n$ through x' of dimension $n - 1$ (or n if x' is an extremum point) and $n - p$, respectively.

Contour Plot with Gradient, Tangent, and Normal Space at (2, 1)



Tangent and normal planes connected to optimization

Consider

$$\min f(x) \quad \text{s.t.} \quad \mathbf{a}(x) = 0,$$

and let $x' \in \mathbb{R}^n$ be regular.

The tangent planes

$$T_{x'} f = \{x \in \mathbb{R}^n \mid \nabla f(x')^\top (x - x') = 0\},$$

$$T_{x'} \mathbf{a} = \{x \in \mathbb{R}^n \mid J_{\mathbf{a}}(x')(x - x') = 0\}, \quad J_{\mathbf{a}}(x') = [\nabla a_1(x') \cdots \nabla a_p(x')]^\top \in \mathbb{R}^{p \times n},$$

are then hyperplanes in $\mathbb{R}^n$ through x' of dimension $n - 1$ (or n if x' is an extremum point) and $n - p$, respectively.

Example

The x_2 -axis can be specified as the feasible region by the equality constraint $a(x_1, x_2) = x_1$ or $a'(x_1, x_2) = x_1^2$. However, the tangent space is only define via $a(x)$ (note that all points on the x_2 -axis are regular for $a(x)$ while non are for $a'(x)$).

Tangent and normal planes connected to optimization

Consider

$$\min f(x) \quad \text{s.t.} \quad a(x) = 0,$$

and let $x' \in \mathbb{R}^n$ be regular.

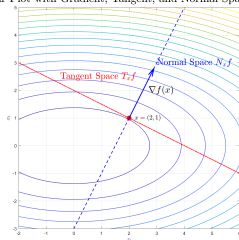
The normal spaces

$$N_{x'} f = \{x \in \mathbb{R}^n \mid x = \alpha \nabla f(x') + x', \alpha \in \mathbb{R}\},$$

$$N_{x'} a = \text{range}(J_a(x')^T) + x' = \text{span}\{\nabla a_1(x), \dots, \nabla a_p(x)\} + x'.$$

are hyperplanes in $\mathbb{R}^n$ through x' of dimension 1 (or 0 if x' is an extremum point) and p , respectively.

Contour Plot with Gradient, Tangent, and Normal Space at (2, 1)



Tangent and normal planes connected to optimization

Example

Consider

$$\min f(x) = x_1^2 + x_2^2 \quad \text{s.t.} \quad \mathbf{a}(x) = -x_2 + x_1^2 + 1 = 0.$$

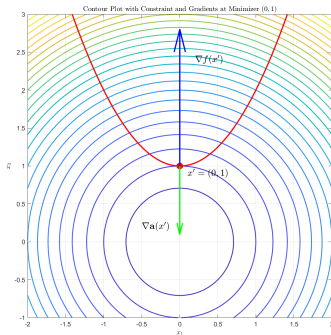
By inspection (e.g., of level sets), it follows that $\mathbf{x}' = [0 \ 1]^\top$ is a minimizer, and that

$$\nabla f(\mathbf{x}') = -2\nabla \mathbf{a}(\mathbf{x}').$$

That is, at the minimizer $\mathbf{x}'$

$$N_{\mathbf{x}'} f = N_{\mathbf{x}'} \mathbf{a}.$$

Something similar is true in general.



First order necessary conditions (equality constraints)

Theorem

Let $x' \in \mathbb{R}^n$ be a regular minimizer for

$$\min f(x) \quad \text{s.t.} \quad \mathbf{a}(x) = 0.$$

Then there exists a vector $\lambda' = [\lambda'_1 \cdots \lambda'_p]^\top$ in $\mathbb{R}^p$ such that

$$\nabla f(x') = J_{\mathbf{a}}(x')^\top \lambda' = \sum \lambda'_i \nabla a_i(x').$$

The scalars λ'_i are called Lagrange multipliers.

Hence at a regular minimizer x' the normal space $N_{x'} f$ is contained in the normal space $N_{x'} \mathbf{a}$.

Note also that minimizer candidates may (in principle) be found by solving the (usually nonlinear) system

$$\begin{aligned} \nabla f(x) - J_{\mathbf{a}}(x)^\top \lambda &= 0, \\ \mathbf{a}(x) &= 0, \end{aligned}$$

of $n + p$ equations in the $n + p$ unknowns (x, λ) .

Example

Consider

$$\min f(x) = x_1^2 + x_2^2 + 1/4x_3^2$$

subject to

$$a_1(x) = -x_1 + x_3 - 1, \quad a_2(x) = x_1^2 + x_2^2 - 2x_1.$$

Hence

$$\mathbf{a}(x) = \begin{bmatrix} -x_1 + x_3 - 1 \\ x_1^2 + x_2^2 - 2x_1 \end{bmatrix} \quad J_{\mathbf{a}}(x)^{\top} = \begin{bmatrix} -1 & 2x_1 - 2 \\ 0 & 2x_2 \\ 1 & 0 \end{bmatrix} \quad \nabla f(x) = \begin{bmatrix} 2x_1 \\ 2x_2 \\ 1/2x_3 \end{bmatrix}.$$

So the equations to be solved

$$\begin{aligned} \nabla f(x) - J_{\mathbf{a}}(x)^{\top} \lambda &= 0, \\ \mathbf{a}(x) &= 0, \end{aligned}$$

are explicit given as

$$2x_1 + \lambda_1 - \lambda_2(2x_1 - 2) = 0$$

$$2x_2 - 2\lambda_2x_2 = 0$$

$$1/2x_3 - \lambda_1 = 0$$

$$-x_1 + x_3 - 1 = 0$$

$$x_1^2 + x_2^2 - 2x_1 = 0$$

Example

Consider

$$\min f(x) = x_1^2 + x_2^2 + 1/4x_3^2$$

subject to

$$a_1(x) = -x_1 + x_3 - 1, \quad a_2(x) = x_1^2 + x_2^2 - 2x_1.$$

So the equations to be solved

$$\begin{aligned}\nabla f(x) - J_a(x)^T \lambda &= 0, \\ \mathbf{a}(x) &= 0,\end{aligned}$$

are explicit given as

$$\begin{aligned}2x_1 + \lambda_1 - \lambda_2(2x_1 - 2) &= 0 \\ 2x_2 - 2\lambda_2x_2 &= 0 \\ 1/2x_3 - \lambda_1 &= 0 \\ -x_1 + x_3 - 1 &= 0 \\ x_1^2 + x_2^2 - 2x_1 &= 0\end{aligned}$$

which yields the two solutions

$$[x' \ \lambda']^T = [0 \ 0 \ 1 \ \frac{1}{2} \ -\frac{1}{4}]^T \quad \text{and} \quad [x', \lambda']^T = [2 \ 0 \ 3 \ \frac{3}{2} \ \frac{11}{4}]^T$$

The Lagrangian

Theorem

Let $x' \in \mathbb{R}^n$ be a regular minimizer for

$$\min f(x) \quad \text{s.t.} \quad \mathbf{a}(x) = 0.$$

Then there exists a vector $\lambda' = [\lambda'_1 \cdots \lambda'_p]^\top$ in $\mathbb{R}^p$ such that

$$\nabla f(x') = J_{\mathbf{a}}(x')^\top \lambda' = \sum \lambda'_i \nabla a_i(x').$$

We may relate the constrained optimization problem in the theorem to an unconstrained optimization problem by introducing the Lagrangian L by

$$L(x, \lambda) = f(x) - \lambda^\top \mathbf{a}(x).$$

Observe that a solution (x', λ') to the unconstrained problem

$$\min L(x, \lambda),$$

has to satisfy (the Fermat's rule)

$$0 = \nabla L(x, \lambda) = \begin{bmatrix} \frac{\partial L}{\partial x}(x, \lambda) \\ \frac{\partial L}{\partial \lambda}(x, \lambda) \end{bmatrix} = \begin{bmatrix} \nabla f(x) - J_{\mathbf{a}}(x)^\top \lambda \\ \mathbf{a}(x) \end{bmatrix}$$

Hence by the theorem x' is also a minimizer candidate for the constrained optimization problem.

First order necessary conditions (general case)

Theorem (Karush-Kuhn-Tucker condition)

Let $x' \in \mathbb{R}^n$ be a regular minimizer for

$$\min f(x) \quad \text{s.t.} \quad a(x) = 0, \quad c(x) \geq_e 0,$$

Then there exists a vector λ' in $\mathbb{R}^p$ and a vector $\mu' \geq_e 0$ in $\mathbb{R}^q$ such that

$$(1) \quad \nabla f(x') = J_a(x')^\top \lambda' + J_c(x')^\top \mu' = \sum \lambda'_i \nabla a_i(x') + \sum \mu'_i \nabla c_i(x'),$$

$$(2) \quad 0 = c(x')^\top \mu' = \sum \mu'_i c_i(x').$$

The scalars λ'_i and μ'_i are called Lagrange (or Karush-Kuhn-Tucker) multipliers.

Note that $c(x)^\top \mu$ in (2), is equivalent to $c_i(x)\mu_i = 0$ for all i . That is, $\mu_i = 0$ if the constraint $c_i(x) \geq 0$ is inactive ($c_i(x) > 0$). So the only terms in (1) which contributes are those connected to active constraints:

$$\nabla f(x') = \sum \lambda'_i \nabla a_i(x') + \sum_{i: c_i(x')=0} \mu'_i \nabla c_i(x').$$

Hence if $\nabla f(x') = 0$ then all multipliers are zero.

First order necessary conditions (general case)

Theorem (Karush-Kuhn-Tucker condition)

Let $x' \in \mathbb{R}^n$ be a regular minimizer for

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Then there exists a vector λ' in $\mathbb{R}^p$ and a vector $\mu' \geq_e 0$ in $\mathbb{R}^q$ such that

$$(1) \quad \nabla f(x') = J_a(x')^\top \lambda' + J_a(x')^\top \mu' = \sum \lambda'_i \nabla a_i(x') + \sum \mu'_i \nabla c_i(x'),$$

$$(2) \quad 0 = c(x')^\top \mu' = \sum \mu'_i c_i(x').$$

The scalars λ'_i and μ'_i are called Lagrange (or Karush-Kuhn-Tucker) multipliers.

If the optimization problem is convex^a then the above necessary conditions are also sufficient (for a global minimizer).

^athat is, if f and $-c$ are convex and $a(x) = Ax - b$ is affine linear

First order necessary conditions (general case)

Theorem (Karush-Kuhn-Tucker condition)

Let $x' \in \mathbb{R}^n$ be a regular minimizer for

$$\min f(x) \quad \text{s.t.} \quad a(x) = 0, \quad c(x) \geq_e 0,$$

Then there exists a vector λ' in $\mathbb{R}^p$ and a vector $\mu' \geq_e 0$ in $\mathbb{R}^q$ such that

$$(1) \quad \nabla f(x') = J_a(x')^\top \lambda' + J_c(x')^\top \mu' = \sum \lambda'_i \nabla a_i(x') + \sum \mu'_i \nabla c_i(x'),$$

$$(2) \quad 0 = c(x')^\top \mu' = \sum \mu'_i c_i(x').$$

The scalars λ'_i and μ'_i are called Lagrange (or Karush-Kuhn-Tucker) multipliers.

Note also that minimizer candidates may (in principle) be found by solving the (usually nonlinear) system

$$\nabla f(x) - J_a(x)^\top \lambda - J_c(x)^\top \mu = 0,$$

$$a(x) = 0,$$

$$c(x)^\top \mu = 0,$$

of $n + p + q$ equations, and the system

$$c(x) \geq_e 0,$$

$$\mu \geq_e 0,$$

of $2q$ inequalities in the $n + p + q$ unknowns (x, λ, μ) .

Second order necessary conditions

Theorem

Let $x' \in \mathbb{R}^n$ be a regular minimizer for

$$\min f(x) \quad \text{s.t.} \quad a(x) = 0, \quad c(x) \geq_e 0,$$

Then there exists a vector λ' in $\mathbb{R}^p$ and a vector $\mu' \geq_e 0$ in $\mathbb{R}^q$ such that

$$N(x')^\top \nabla_x^2 L(x', \lambda', \mu') N(x') \geq 0 \quad (\text{positive semi-definite})$$

where

- Lagrangian:

$$L(x, \lambda, \mu) = f(x) - \sum \lambda_i a_i(x) - \sum \mu_i c_i(x).$$

- The Hessian of L with respect to x :

$$\nabla_x^2 L(x, \lambda, \mu) = \left[\frac{\partial^2 L}{\partial x_j \partial x_i}(x, \lambda, \mu) \right] = H_f(x) - \sum \lambda_i H_{a_i}(x) - \sum \mu_i H_{c_i}(x),$$

with $H_f(x)$, $H_{a_i}(x)$ and $H_{c_i}(x)$ the Hessian of f , a_i and c_i at x , respectively.

- $N(x)$ is a matrix whose columns form a basis for the null space of

$$\begin{bmatrix} J_a(x) \\ J_c(x) \end{bmatrix}.$$

Second order sufficient conditions

Theorem

A point $x' \in \mathbb{R}^n$ is a strong local minimizer for

$$\min f(x) \quad \text{s.t.} \quad a(x) = 0, \quad c(x) \geq_e 0,$$

if x' is regular and there exists a vector λ' in $\mathbb{R}^p$ and a vector $\mu' \geq_e 0$ in $\mathbb{R}^q$ such that

$$N(x')^\top \nabla_x^2 L(x', \lambda', \mu') N(x') > 0 \quad (\text{positive definite})$$

where

- Lagrangian:

$$L(x, \lambda, \mu) = f(x) - \sum \lambda_i a_i(x) - \sum \mu_i c_i(x).$$

- The Hessian of L with respect to x :

$$\nabla_x^2 L(x, \lambda, \mu) = \left[\frac{\partial^2 L}{\partial x_i \partial x_j}(x, \lambda, \mu) \right].$$

- $N(x)$ is a matrix whose columns form a basis for the null space of

$$\begin{bmatrix} J_a(x) \\ J'_c(x) \end{bmatrix},$$

with $J'_c(x)$ the “active” Jacobian of c at x (that is, consist of rows that are composed of those gradients of inequality constraints that are active at x).